

BARATH B

Android Engineer · Kotlin Multiplatform · Full-Stack Backend

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SUMMARY

Built and shipped 3 production apps — Android, Desktop, and backend — as a solo engineer still in college. Specializes in Kotlin Multiplatform and full-stack Android systems, from custom TCP protocols to distributed Spring Boot backends serving 1,200 daily users.

EDUCATION

B.E Computer Science Engineering

Expected 2027

Government College of Engineering Srirangam, Trichy · CGPA: 7.6

Relevant course:

Operating Systems , Data structure , Algorithms , DPCO

SKILLS

Mobile: Jetpack Compose · MVVM/MVI · Clean Architecture · Hilt/Dagger · Coroutines & Flow · Room · Retrofit · DataStore · KMP · Compose Multiplatform · Firebase

Backend: Spring Boot 3.5 · Spring Security · PostgreSQL · Flyway · JPA/Hibernate · JWT · ShedLock · Bucket4j · Caffeine Cache · Docker

DevOps & Tooling: Git · Fastlane · GitHub Actions · AUR packaging · Bash · SQLite · jarsigner · ADB

Languages: Kotlin · C++ · Python

PROJECTS

GCES Hostel Management System · Android · Spring Boot · PostgreSQL · Firebase

Jan 2026 – Mar 2026

End-to-end institutional app — the only meal and hostel management system for a 1,200-student government college. Live on Play Store · 1,200 DAU.

- Multi-module Clean Architecture (:app Android, :server Spring Boot, :common shared DTOs) with Hilt DI scoping ViewModels, repositories, and singleton managers
- Predictive meal engine; ShedLock distributed JDBC locks guarantee atomic midnight state transitions across server instances — zero race conditions at scale
- Caffeine in-memory cache + Bucket4j token-bucket rate limiting absorb rush-hour bursts, holding API response under sub-100ms at peak load
- JWT stateless auth + Spring Security; PostgreSQL with Flyway versioned migrations; deployed on Render (JAR) and HuggingFace Spaces (Docker)
- Firebase FCM push, Crashlytics, Performance Monitoring; custom Circular Reveal theme transition in Compose

URL: <https://play.google.com/store/apps/details?id=com.barath.gceshostel>

ConnectLnx · Kotlin Multiplatform · Compose Multiplatform · mDNS · ML Kit

Dec 2025 – Jan 2026

Fastest way to move files between devices — no cloud, no account, no limits. Ships to Android, Linux, and Windows from a single codebase.

- Custom binary TCP protocol with parallel multi-socket streaming — files chunked across up to 5 concurrent connections for maximum LAN throughput
- Zero-config discovery via mDNS on Android (NsdManager) and Desktop (jmdns); QR code pairing as network-restriction fallback
- Single codebase → 4 distribution targets: Play Store (.aab), Linux (.deb), Windows (.msi via GitHub Actions), AUR (connectlnx-bin)

URL: <https://connectlnx.web.app/>

Aztrox Release · Bash · SQLite · Fastlane · GitHub Actions · jarsigner · gh CLI

Apr 2026

Custom CI/CD engine built because no existing tool understood the full Aztrox stack. Manages 3 live projects across 6 distribution targets.

- Unified CLI handling build → sign → distribute lifecycle across Google Play, GitHub Releases, AUR, Render, HuggingFace Spaces, and Firebase
- LIFO atomic rollback registry — pipeline failures auto-revert version bumps, gradle.properties, and git tags without manual intervention
- Parallel build orchestration: .deb + .aab built locally while a release/v{version} branch triggers GitHub Actions to build .msi simultaneously
- Preflight health system validates dependencies, keystore passwords, .env vars; every release logged to SQLite with auto-generated HTML audit reports